Unit 3 Predicted Exam

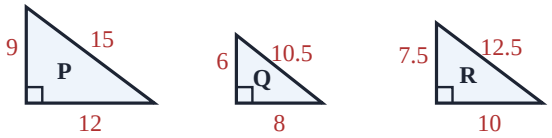
Math 9E Lessons 3.1–3.3 10 questions · 34 marks Name: _____

These are the questions most likely to appear on your test, predicted from the starred problems in your Unit 3 worksheets. Show all your work. Answers are on the last page.

Question 1

[2 marks]

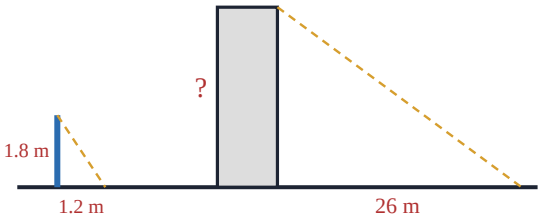
Triangle P has sides 9, 12, 15. Triangle Q has sides 6, 8, 10.5. Triangle R has sides 7.5, 10, 12.5. Which triangle(s) are similar to P? Show the ratios.



Question 2

[3 marks]

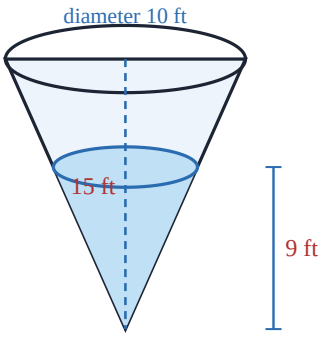
A 1.8 m pole casts a 1.2 m shadow. At the same time a building casts a 26 m shadow. (a) Find the building's height. (b) Later, the building's shadow is 39 m. How long is the pole's shadow then?



Question 3

[3 marks]

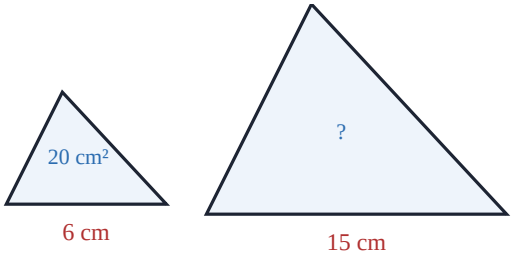
A conical tank (point down) is 15 ft deep with top diameter 10 ft. It is filled to a depth of 9 ft. Find (a) the area of the water surface and (b) the volume of water.



Question 4

[3 marks]

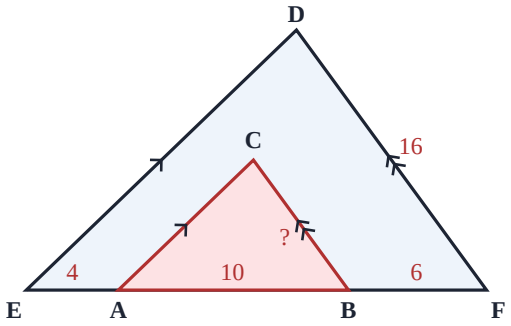
Corresponding sides of two similar triangles are 6 cm and 15 cm. The small triangle has area 20 cm². Find the area of the large one.



Question 5

[4 marks]

In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC ∥ ED and BC ∥ FD. FD = 16. (a) Prove ΔABC ~ ΔEFD. (b) Find BC.



Question 6

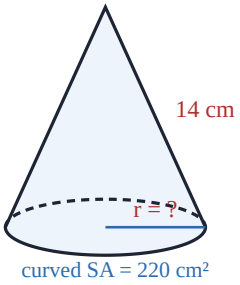
[2 marks]

A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

Question 7

[3 marks]

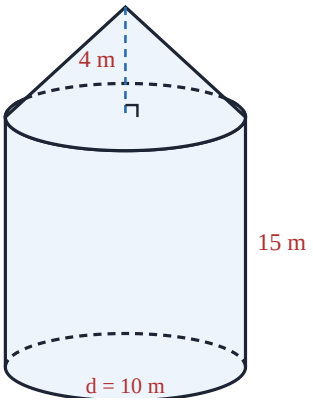
A cone has curved surface area 220 cm^2 and slant height 14 cm. Find r to 1 decimal place.



Question 8

[5 marks]

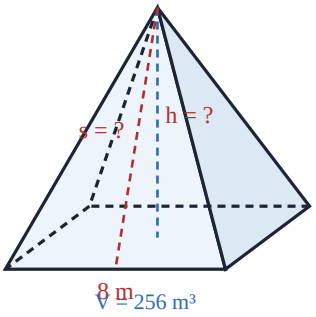
A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 m^2 and costs \$65. Find the area (2 decimal places), number of buckets and total cost.



Question 9

[4 marks]

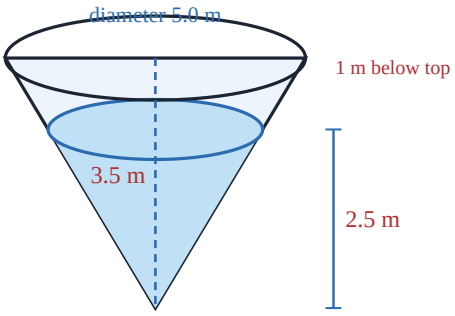
A right square pyramid has volume 256 m^3 and base side 8 m. Find its height and its slant height to the nearest hundredth.



Question 10

[5 marks]

An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?



ANSWER KEY

Q1 P/Q: 9/6 = 1.5, 12/8 = 1.5, 15/10.5 ≈ 1.43 ⇒ **not similar**. P/R: 9/7.5 = 12/10 = 15/12.5 = 1.2 ⇒ **R is similar to P**.

Q2 (a) H/26 = 1.8/1.2 ⇒ **H = 39 m**. (b) Now height = shadow (39 m : 39 m), so the pole's shadow = **1.8 m**.

Q3 r = 5 × 9/15 = 3 ft. (a) $\pi(3^2)$ = **$9\pi \approx 28.27$ ft²**. (b) $\frac{1}{3}\pi(9)(9)$ = **$27\pi \approx 84.82$ ft³**.

Q4 k = 15/6 = 2.5 ⇒ $k^2 = 6.25$. Area = 20 × 6.25 = **125 cm²**.

Q5 (a) AC ∥ ED ⇒ ∠CAB = ∠DEF (corresponding angles). BC ∥ FD ⇒ ∠CBA = ∠DFE. So ΔABC ~ ΔEFD (AA). (b) EF = 4 + 10 + 6 = 20, k = AB/EF = 10/20 = 1/2 ⇒ BC = ½ · FD = **8**.

Q6 0.6 g = 600 mg. 600 × 5/250 = **12 mL = 2.4 tsp**.

Q7 $\pi r(14) = 220$ ⇒ $r = 220/(14\pi) \approx$ **5.0 cm**.

Q8 r = 5. Roof slant = $\sqrt{(5^2 + 4^2)} = \sqrt{41} \approx 6.403$. Walls $2\pi(5)(15) = 150\pi$; roof $\pi(5)(6.403) \approx 32.02\pi$. Total ≈ **571.82 m²**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

Q9 $\frac{1}{3}(64)h = 256$ ⇒ **h = 12 m**. Slant = $\sqrt{(12^2 + 4^2)} = \sqrt{160} \approx$ **12.65 m**.

Q10 (a) $V = \frac{1}{3}\pi(2.5^2)(3.5) \approx 22.91$ m³ ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ k = 2.5/3.5 = 5/7. V = $22.91 \times (5/7)^3 \approx$ **8.3 kL**. (c) $(5/7)^3 \approx$ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

WHERE EACH QUESTION COMES FROM

Q1	Practice 3.1 #1 (are the triangles similar?)
Q2	Example 3 and Practice 3.1 #4 (flagpole, Delbert)
Q3	Practice 3.1 #5 (conical tank)
Q4	Practice 3.1 #7 ★★★+ (area ratio)
Q5	Practice 3.1 #9 (prove similar, find x)
Q6	Practice 3.2 #3–4 (medicine dose)
Q7	Practice 3.2 #7 (cone radius from curved SA)
Q8	Practice 3.2 #8 ★★★ (silo sealant)
Q9	Practice 3.3 #5 ★★★ (pyramid slant from volume)
Q10	Practice 3.3 #4b ★★★+ (tank 1 m below top)